



Performance Tuning and Monitoring **Part 3**

Here're some tricks to help you counter disk fragmentation and improve RAID performance on hard drives significantly.

In the October issue of LFY, we discussed the different performance tuning methods related to hard disk drives, using IO elevators. We had also explored the queuing theory and finding hotspots using the `strace` command. Now, after our Diwali break, it's time we resume fine tuning our hard disks again.

This time we will look at how to counter disk fragmentation and improve RAID performance significantly. We will also learn about cache profiling using `Valgrind`.

Countering disk fragmentation

Disk fragmentation affects the sequential read access performance. This is because it results in extra head movement on the disk drive. In fact, the Linux filesystem is designed to minimise the effects of fragmentation as much as possible. When a file is created or extended, the filesystem always tries to allocate blocks for the file from the same block group that contains its inode. And to ensure that files are spread evenly in the

partition, when a new file is created, an attempt is made to place it in some block group other than the one containing its parent directory.

To view the fragmentation of a file we can use the following command:

```
$ filefrag -v /path/to/the/filename
```

To check the fragmentation on a mounted filesystem, issue the command shown below:

```
$ dumpe2fs /dev/sdaX
```

...where X is the partition number.

Types of journaling in ext3

One major issue in journaling filesystems is whether they only log changes to the filesystem metadata or log changes to all filesystem data, including changes to the files themselves. The ext3 filesystem supports three different journaling modes. Here's how Wikipedia [<http://en.wikipedia>].